

VARANASI DISTRICT — VILLAGE ENVIRONMENT & CARBON ASSESSMENT

M.Tech Thesis Project: Village-Level Resource Mapping & Greenhouse Gas Accounting

Village: Ausanpur	Census Code: 208818	Block: Harahua
Tehsil / Sub-District: Pindra	District / State: Varanasi, UP	Coordinates: 25.3813° N, 82.8660° E

1. DEMOGRAPHY & LAND USE INVENTORY

Demographic Parameter	Value	Land Use Parameter	Area (Hectares)
Total Residents	3,077	Total Geographical Area	230.00 ha
Total Households	478	Settlement / Built-up Area	24.00 ha
Male Residents	1,578	Agricultural / Net Sown Area	189.00 ha
Female Residents	1,499	Forest & Plantation Area	0.00 ha
Average Household Size	6.4 persons	Remaining / Other Land	17.00 ha

2. WATER, ENERGY, TRANSPORT & SOLID WASTE ACTIVITY DATA

Resource / Activity Domain	Metric	Observed Survey Value	Annual Total
Water Resources: Ponds & Tanks	Surface Area	0.00 ha	-
Water Resources: Canals & Drainage	Surface Area	0.00 ha	-
Water Resources: Rivers & Water Bodies	Total Water Area	0.00 ha	-
Domestic Energy: LPG	Consumption (Monthly)	1,625.2 kg/month	19,502.4 kg/yr
Domestic Energy: Firewood / Biomass	Consumption (Monthly)	8,699.6 kg/month	104,395.2 kg/yr
Domestic Energy: Grid Electricity	Consumption (Monthly)	17,447.0 kWh/month	209,364.0 kWh/yr
Road Transport: Motorbikes (2-W)	Vehicle Count & Petrol	153 units (1,499.4 L/mo)	17,992.8 L/yr
Waste Management: Rural Solid Waste	Waste Generation	17,846.6 kg/month	214,159.2 kg/yr

3. LIVESTOCK POPULATION INVENTORY

Livestock Category	Head Count	IPCC Enteric + Manure Factor	Estimated Annual GHG
Dairy Cattle & Cows	229 heads	980.0 kg CO2e / head / year	224.42 t CO2e/yr
Goats (Caprine)	167 heads	145.6 kg CO2e / head / year	24.32 t CO2e/yr
Sheep (Ovine)	38 heads	145.6 kg CO2e / head / year	5.53 t CO2e/yr
Total Livestock	434 heads	-	254.27 t CO2e/yr

4. GREENHOUSE GAS ACCOUNTING & EMISSION MODELING RESULTS

Emission Source / Sector	Gas Species	Emission Factor & Standard	Annual Output (t CO2e)	Share (%)
LPG Domestic Combustion	CO2, CH4, N2O	2.984 kg CO2e/kg (IPCC 2006)	58.20 t	7.2%
Firewood Biomass Burning	CO2, CH4, N2O	1.747 kg CO2e/kg (MoEFCC/IPCC)	182.38 t	22.7%
Electricity Grid Consumption	CO2	0.820 kg CO2/kWh (CEA India v20)	171.68 t	21.3%
2-Wheeler Petrol Transport	CO2, CH4, N2O	2.310 kg CO2e/L (IPCC 2006)	41.56 t	5.2%

Rural Solid Waste Disposal	CH4	0.450 kg CO2e/kg (IPCC FOD)	96.37 t	12.0%
Livestock Enteric & Manure	CH4	IPCC Tier 1 Ruminant Factors	254.27 t	31.6%
TOTAL VILLAGE EMISSION	All GHGs	Aggregated Carbon Footprint	804.46 t CO2e/yr	100.0%
PER CAPITA CARBON FOOTPRINT	CO2 Equivalent	Normalized by Population	261.44 kg/capita/yr	-

5. SCIENTIFIC METHODOLOGY & CITATION STATEMENT

Methodological Distinction: Survey parameters (demography, land area, fuel volumes, vehicle counts, and livestock head counts) represent direct primary/secondary survey data from Census 2011 and field research. Greenhouse gas emissions are derived model outputs computed via Tier-1 emission factor multiplication: *Emissions (t CO2e) = Activity Data × EF × 10⁻³*.

References:

- Central Electricity Authority (CEA), Government of India (2024): *CO2 Baseline Database for the Indian Power Sector, User Guide Version 20.0*.
- Intergovernmental Panel on Climate Change (IPCC) (2006): *IPCC Guidelines for National Greenhouse Gas Inventories*, Vol 2 (Energy), Vol 4 (AFOLU), Vol 5 (Waste).
- Ministry of Environment, Forest and Climate Change (MoEFCC), India: *India's Biennial Update Reports to UNFCCC*.